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See application file for complete search history.

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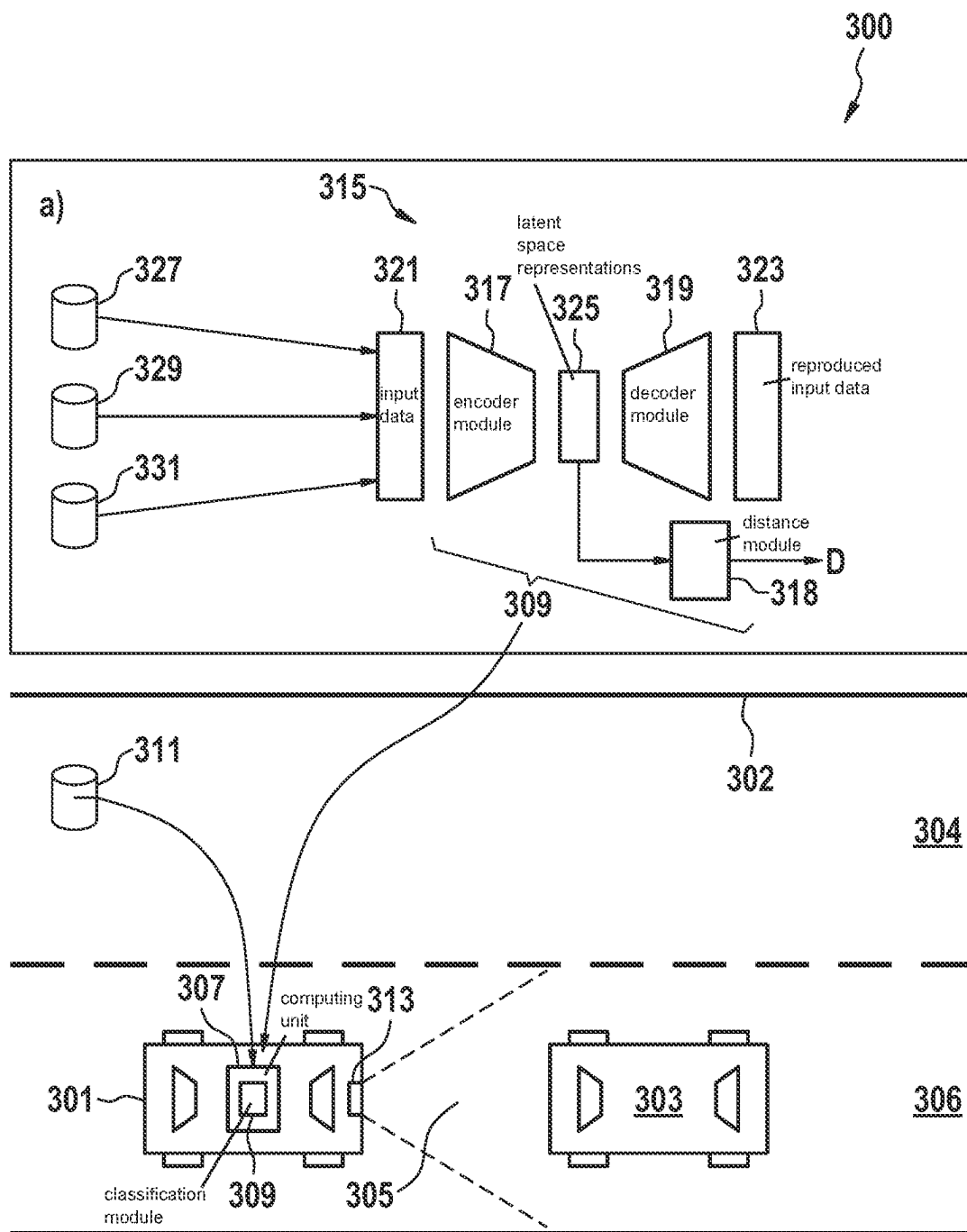


FIG. 1

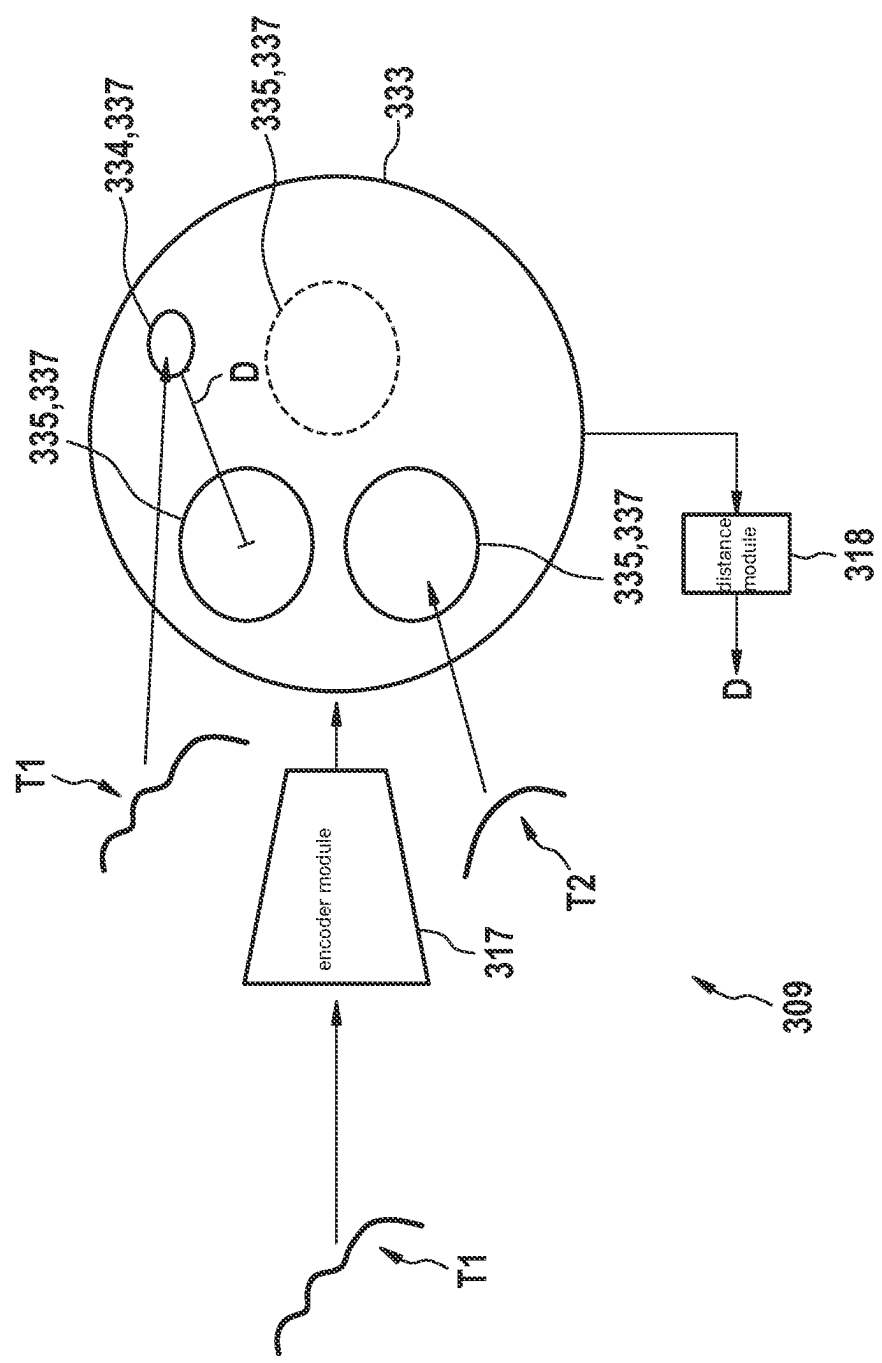


FIG. 2

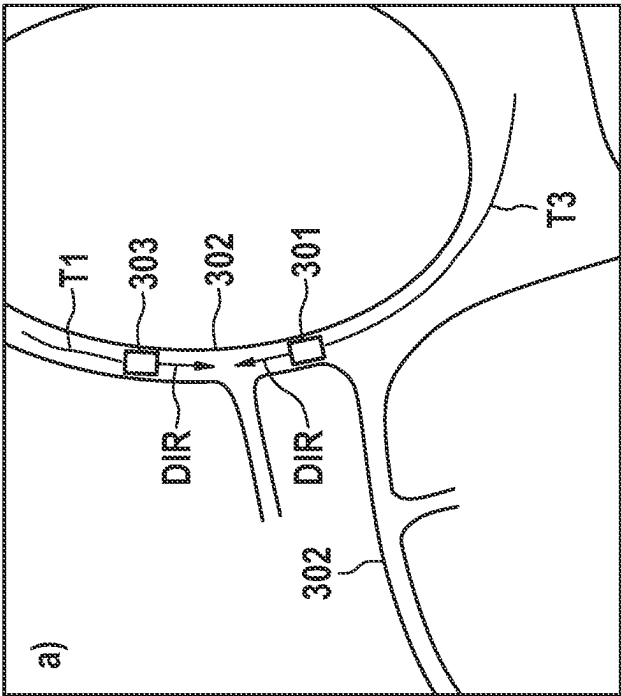
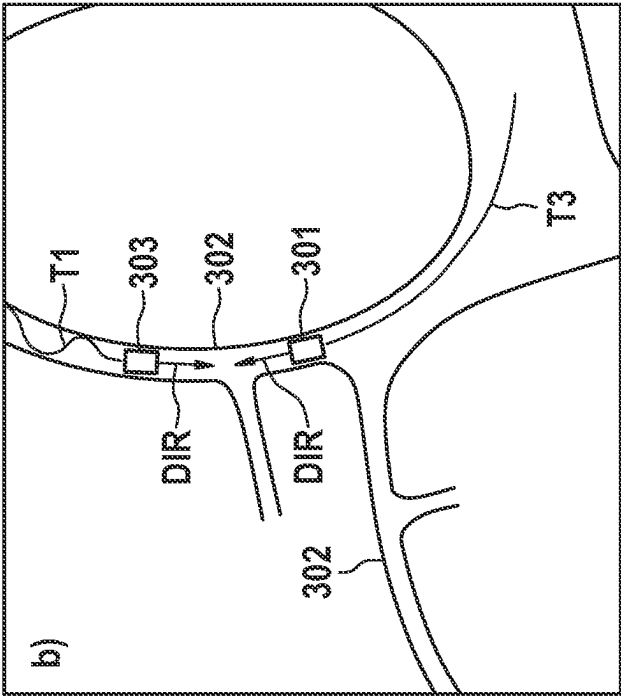


FIG. 3

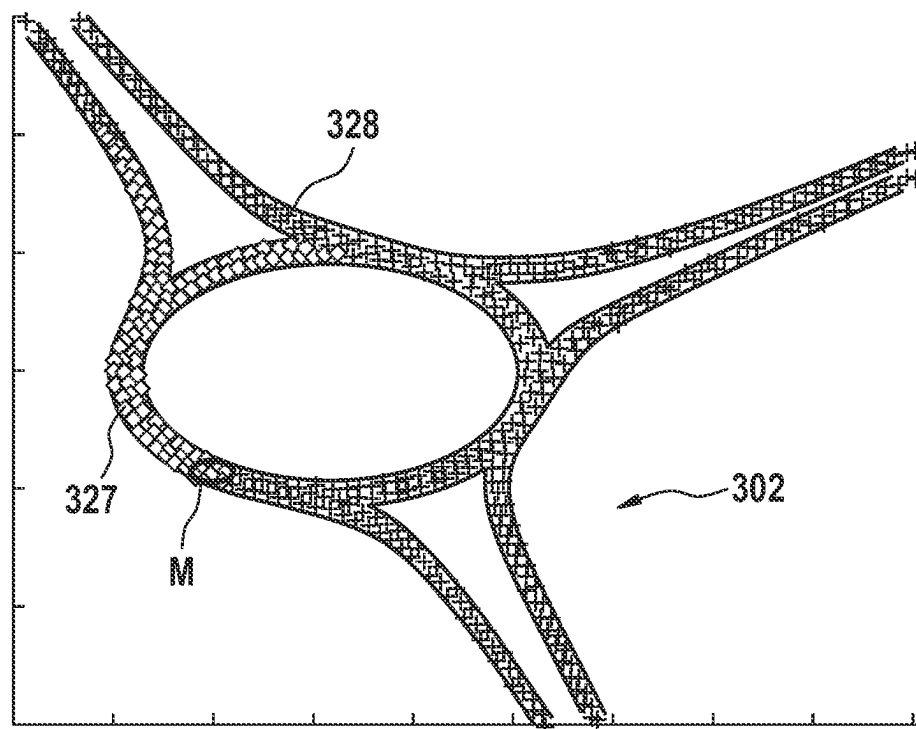


FIG. 4

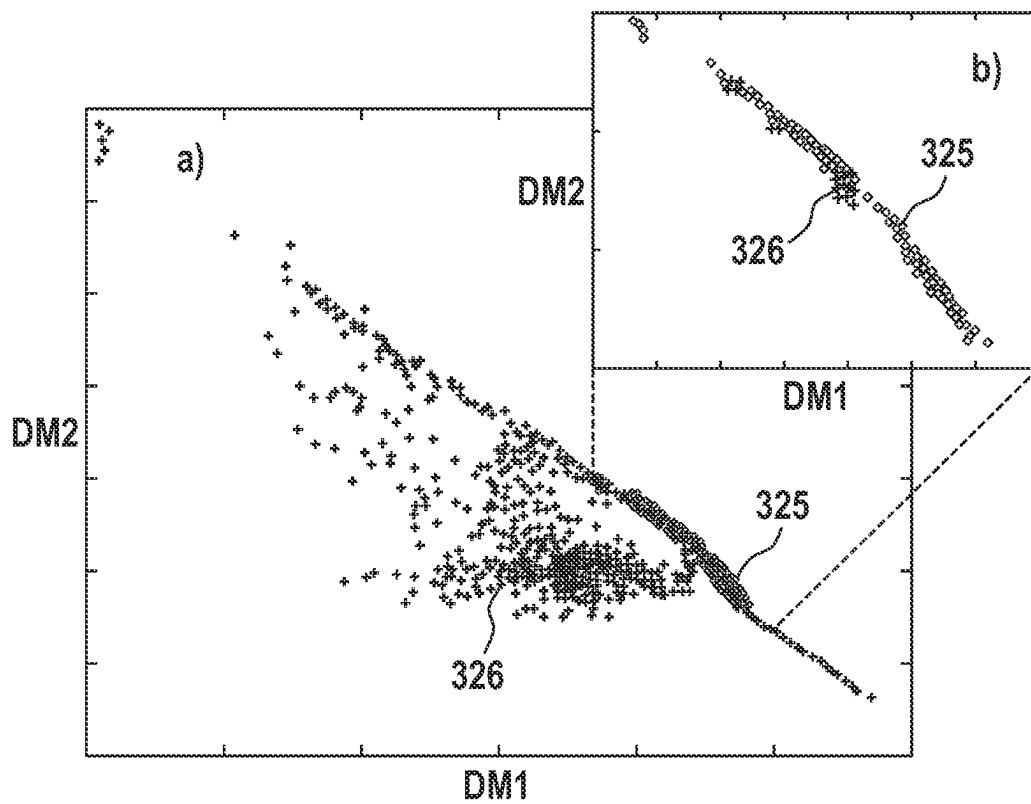


FIG. 5

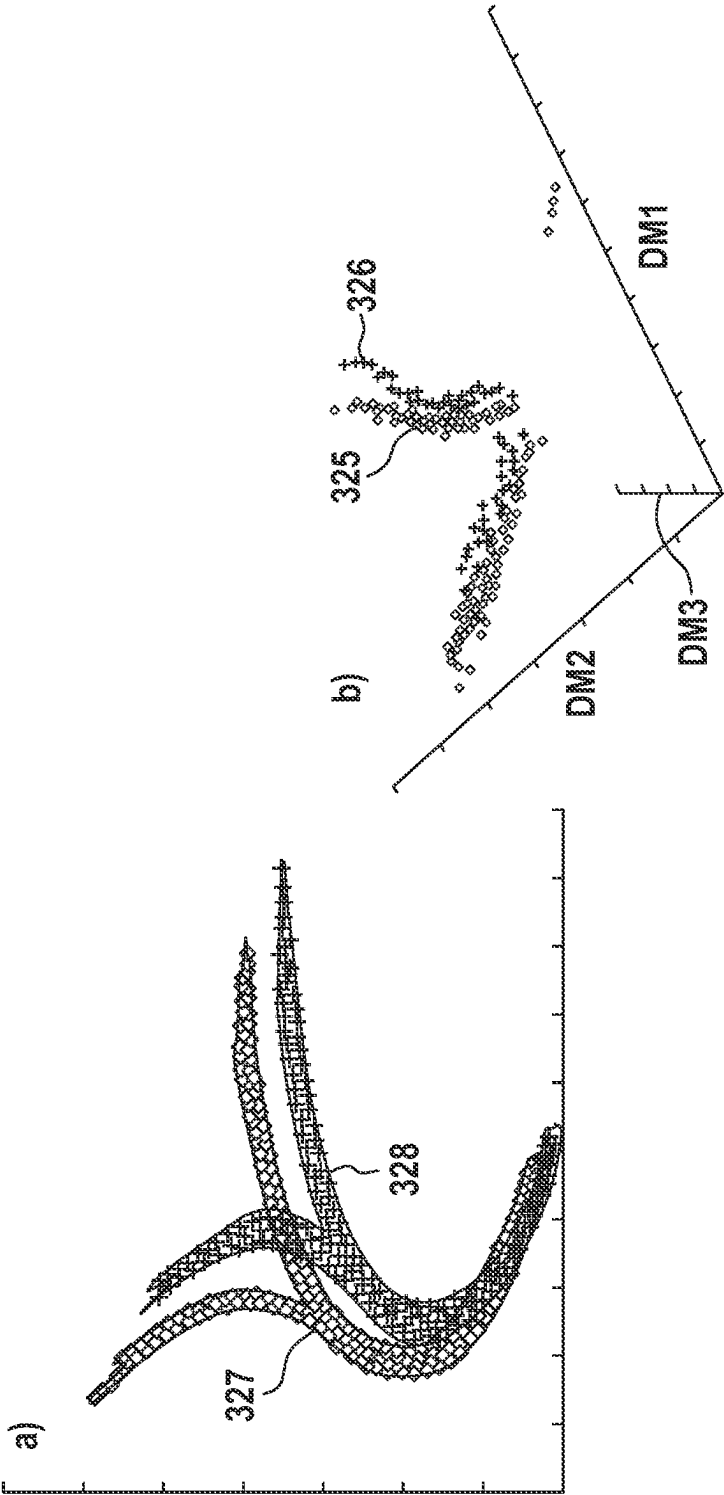


FIG. 6

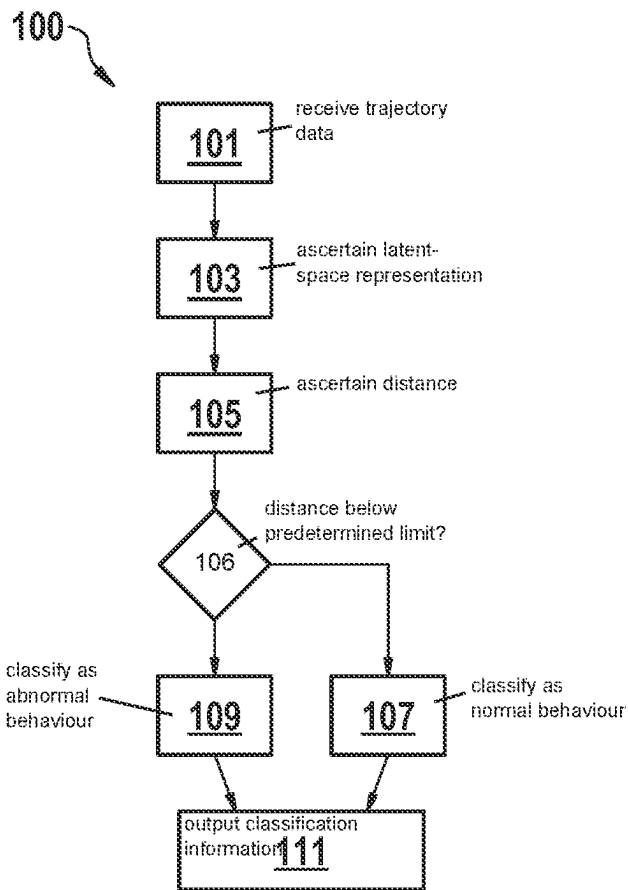


FIG. 7

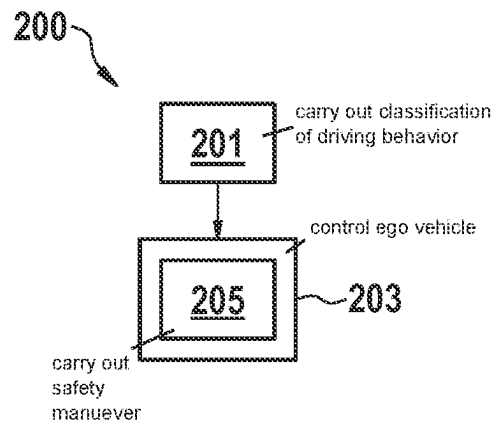


FIG. 8

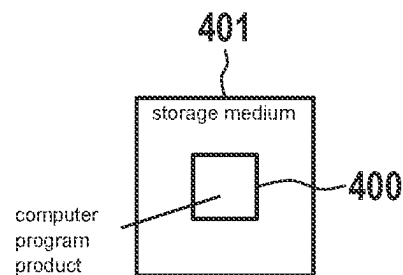


FIG. 9

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METHOD FOR CLASSIFYING A BEHAVIOR OF A ROAD USER AND METHOD FOR CONTROLLING AN EGO VEHICLE

CROSS REFERENCE

The present application claims the benefit under 35 U.S.C. § 119 of German Patent Application No. DE 10 2022 205 100.9 filed on May 23, 2022, which is expressly incorporated herein by reference in its entirety.

FIELD

The present invention relates to a method for classifying a driving behavior of a road user in an environment of an ego vehicle. The present invention also relates to a method for controlling an ego vehicle.

BACKGROUND INFORMATION

A proper assessment of future driving behavior of further road users is essential for controlling vehicles, in particular autonomously controllable vehicles. For safely controlling the vehicle, it is in particular important to be able to distinguish between normal driving behavior, where unforeseeable actions are not expected, and abnormal driving behavior, which deviates from normal driving behavior and where unforeseeable actions are expected to be carried out in the future. Such recognition of abnormal behavior enables early adjustment of the driving behavior of the vehicle to be controlled.

SUMMARY

It is an object of the present invention to provide an improved method for classifying a driving behavior of a road user in an environment of an ego vehicle and an improved method for controlling an ego vehicle.

This object may be achieved by the method for classifying a driving behavior of a road user in an environment of an ego vehicle and by the method for controlling an ego vehicle according to the present invention. Advantageous configurations of the present invention are disclosed herein.

According to an aspect of the present invention, a method for classifying a driving behavior of a road user in the environment of an ego vehicle is provided. According to an example embodiment of the present invention, the method comprises:

receiving trajectory data of a driving trajectory of a road user arranged in the environment of an ego vehicle, wherein the trajectory data comprise at least position information of the driving trajectory followed by the road user;

ascertaining a latent-space representation of the driving trajectory of the road user in a latent space, wherein the latent-space representation of the driving trajectory is given by a reduced-dimension representation of the trajectory data;

ascertaining a distance of the latent-space representation of the driving trajectory of the road user to a latent-space representation of at least one normal trajectory in the latent space, wherein the normal trajectory is classified as a driving trajectory based on a normal driving behavior; and

classifying a driving behavior of the road user as a normal driving behavior if the distance of the latent-space representation of the driving trajectory of the road user

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to the latent-space representation of the at least one normal trajectory in the latent space falls below a predetermined limit value; or

classifying the driving behavior of the road user as an abnormal driving behavior if the distance of the latent-space representation of the driving trajectory of the road user to the latent-space representation of the at least one normal trajectory in the latent space exceeds the predetermined limit value; and

outputting classification information.

This can achieve the technical advantage that an improved method for classifying a driving behavior of a road user in the environment of an ego vehicle can be provided. For this purpose, according to an example embodiment of the present invention, trajectory data of a driving trajectory of the road user are received and latent-space representations for the corresponding trajectory data are ascertained. The latent-space representations are representations of the trajectory data in the latent space and describe a reduced-dimension representation of the trajectory data.

According to an example embodiment of the present invention, corresponding representations in the latent space are calculated by an appropriately trained artificial intelligence based on the received trajectory data of the driving trajectories of the road users. For example, these representations may be designed in vector representation and represent reduced-dimension descriptions of the information of the trajectory data. The conversion of the trajectory data of the driving trajectories into corresponding latent-space representations can thus be interpreted as an encoding of the information of the trajectory data. The latent-space representations comprise the information or the majority of the information of the respectively associated trajectory data but have a reduced-dimension representation in comparison to the representation of the original trajectory data. The latent space is here seen as the set of all latent-space representations of the trajectory data and can be multi-dimensional according to the dimensionality of the latent-space representations.

The present invention is now based on the following idea that, when converting the trajectory data into corresponding latent-space representations, the appropriately trained artificial intelligence performs this calculation of the latent-space representations according to relevant criteria and similar or identical trajectory data thus result in similar or identical latent-space representations. On the other hand, different trajectory data, which are based on very different driving trajectories, will have latent-space representations that differ greatly from one another. Latent-space representations within the meaning of the application are designed as vector representations. Identical or similar latent-space representations are therefore vector representations with identical or only slightly different entries. Slightly different entries result in small differences, while the same entries have a difference of zero. Since similar or identical vector representations result in small or vanishing differences, the similar or identical latent-space representations will thus be arranged very closely together within the latent space. On the other hand, greatly different trajectory data, which are based on very different driving trajectories, result in latent-space representations that are arranged at a greater distance from one another in the latent space.

Since the latent-space representations are preferably designed in vector representation, a similarity determination of the respectively associated trajectory data or of the corresponding driving trajectories can thus be ascertained via a simple determination of the distance between the

individual latent-space representations within the latent space. Similar or identical driving trajectories thus result in latent-space representations with very small distances in the latent space, while greatly different driving trajectories have latent-space representations that have correspondingly larger distances from one another in the latent space.

Furthermore, the present invention is based on the idea that driving trajectories based on a normal driving behavior will be greatly similar to one another, while driving trajectories based on an abnormal driving behavior will have great differences from the driving trajectories of the normal driving behavior.

A normal driving behavior is characterized in that traffic regulations are complied with. The respective vehicle thus drives in the prescribed direction of travel in the correspondingly designated lane of the roadway. No unforeseen driving maneuvers are carried out. The vehicle instead follows the path of the roadway evenly, without jerky steering movements and with an even velocity profile. Furthermore, the vehicle only drives in the lanes provided for this purpose, and not beyond roadway boundaries or, for example, on the walkway.

Driving trajectories for vehicles driving in this normal manner will thus be highly similar to one another since the paths of the respective driving trajectories are largely defined by the paths of the roadways traveled and by the prevailing traffic rules.

On the other hand, vehicles that do not adhere to the path of the roadway or to the prevailing traffic rules have driving trajectories that are highly different from the driving trajectories of the normal driving behavior. These driving trajectories may have any deviations from the above-described behaviors of vehicles driving normally. For example, the trajectories may have deviations from the paths of the roadways traveled. Jerky steering movements or wavy lines or zig-zag movements may be carried out. Great velocity differences or the general non-compliance with prevailing traffic rules may be discernible. The respective vehicles may deviate from the intended lanes and may drive on walkways or in the wrong direction of travel, for example.

While driving trajectories of a normal driving behavior thus have identical or very similar latent-space representations that are arranged very closely together in the latent space, driving trajectories based on an abnormal driving behavior have latent-space representations that are greatly different from the latent-space representations of the normal driving behavior and are arranged correspondingly far from these latent-space representations in the latent space.

Based on this idea, the method according to an example embodiment of the present invention provides to classify the driving behavior of a road user in the environment of the ego vehicle based on the respective driving trajectory of the road user by determining the distance of a latent-space representation of the trajectory data of the driving trajectory of the road user to known latent-space representations of driving trajectories that are known to be based on a normal driving behavior and that are referred to in the present application as normal trajectories, as normal driving behavior or as abnormal driving behavior.

The determination of the distance of the latent-space representations of the driving trajectories of the road users in the environment of the ego vehicle to the latent-space representations of the normal trajectories, based on a normal driving behavior, in the latent space thus enables a simple and precise classification of the driving behavior of the road user in the environment of the ego vehicle. The driving behavior of the road user in the environment of the ego

vehicle is classified as a normal driving behavior if the respective latent-space representation of the driving trajectory of the road user has a distance to latent-space representations of normal trajectories in the latent space that is less than a predetermined limit value. If, on the other hand, the distance between the latent-space representation of the driving trajectory of the road user to the latent-space representations of the normal trajectories in the latent space is greater than the predetermined limit value, the driving behavior of the road user, which has led to the respective driving trajectory, is classified as an abnormal driving behavior.

Based on the conducted classification of the driving behavior of the road users in the environment of the ego vehicle, a control of the ego vehicle can be adjusted to the correspondingly classified behavior of the road user. For example, safety functions predetermined by the ego vehicle may be carried out if an abnormal driving behavior is detected. This can in turn improve the control of the ego vehicle and can increase the safety of the ego vehicle.

The ego vehicle may be a partially automated, highly automated or fully automated controllable vehicle within the meaning of the application. Road users may include further vehicles, in particular trucks, passenger cars, buses, motorcycles, or bicycles. A roadway within the meaning of the application may be a highway, a country road, an inner-city road, a traffic-calmed road, or a parking lot.

According to one embodiment of the present invention, the distance is determined by a predetermined distance metric, wherein the distance metric is given by a Euclidean distance function or by a logarithmic likelihood function of the latent-space representations within the latent space.

This can achieve the technical advantage that a precise distance determination between the latent-space representations of the trajectory data of the driving trajectories of the road users in the environment of the ego vehicle and the latent-space representations of the known normal trajectories of the known normal driving behavior within the latent space is enabled. Since the latent-space representations are preferably designed as vector representations, it is possible to perform technically simple distance determinations between various latent-space representations. This enables precise distance determination and, based thereon, precise classification of the respective driving behavior.

According to one embodiment of the present invention, the classification is carried out by a classification module, wherein the classification module comprises an appropriately trained encoder module of an autoencoder architecture for ascertaining the latent-space representation of the driving trajectory of the road user, wherein the autoencoder architecture is trained to ascertain latent-space representations of driving trajectories in the latent space and to recognize and classify them as driving trajectories based on normal driving behavior.

This can achieve the technical advantage that precise classification of the driving behavior of the road users is enabled. For this purpose, an autoencoder architecture is used to learn the latent-space representations of the trajectory data, in particular of the normal trajectories, during training of the autoencoder architecture. The corresponding encoder module of the autoencoder architecture is correspondingly configured, upon completion of the training, to generate latent-space representations of trajectory data of driving trajectories. As described above, the latent-space representations are calculated based on learned criteria so that similar or identical trajectory data of similar or identical driving trajectories results in similar or identical latent-space

representations. Trajectory data of trajectories that deviate greatly from normal trajectories, on the other hand, result in greatly deviating or different latent-space representations. Through the corresponding training of the autoencoder architecture, the respective encoder module is thus configured to arrange latent-space representations of similar driving trajectories in the latent space closely together and to arrange latent-space representations of greatly deviating driving trajectories in the latent space correspondingly further away.

Latent-space representations are arranged closely together in the latent space within the meaning of the application if the respective vector representations of the latent-space representations have similar or identical entries. On the other hand, latent-space representations are arranged correspondingly apart from one another in the latent space within the meaning of the application if the respective vector representations of the latent-space representations have greatly different entries from one another.

According to one embodiment of the present invention, the autoencoder architecture is designed as a variational autoencoder VAE, wherein the latent-space representations of the driving trajectories are designed as statistical distributions in the latent space and are each characterized by an expected value μ and a variance σ^2 .

This can achieve the technical advantage that improved trainability of the autoencoder architecture can be achieved. By designing the autoencoder architecture as a variational autoencoder and the latent-space representations as statistical distributions, overfitting of the autoencoder architecture during the training process can be avoided. Through the statistical distributions as latent-space representations, a degree of uncertainty can be integrated into the training of the autoencoder architecture. This forces the autoencoder architecture during the training to learn the underlying mechanisms. This in turn improves the ability to generalize the trained autoencoder architecture, which in turn enables improved performance of the trained autoencoder architecture.

According to one embodiment of the present invention, the distributions in the latent space are designed as Gaussian normal distributions.

This can achieve the technical advantage that a statistical distribution of the latent-space representations that is as simple and easily manageable as possible is achieved. The Gaussian normal distribution facilitates the handling of the statistical distributions of the latent-space representations.

According to one embodiment of the present invention, the training of the autoencoder architecture is carried out on trajectory data of a plurality of trips of a vehicle or of a plurality of vehicles along a roadway traveled by the ego vehicle, wherein map information of a map representation mapping the respective traveled roadway and/or information regarding a roadway condition and/or information regarding prevailing traffic regulations and/or information regarding prevailing traffic volumes and/or information regarding prevailing weather conditions and/or time-of-day and/or day-of-the-week information are also taken into account in the training.

This can achieve the technical advantage that precise training of the autoencoder architecture is enabled. For this purpose, trajectory data recorded on a plurality of trips of a plurality of vehicles or of at least one vehicle along various roadways are taken into account as training data. For example, only trajectory data of driving trajectories based on a normal driving behavior of the respective vehicles may be taken into account in the training of the autoencoder archi-

itecture. During the training, the corresponding autoencoder architecture is thus trained to generate the corresponding latent-space representations of the trajectory data of the normal trajectories and to arrange them in the latent space correspondingly closely to one another due to the high similarity of the normal trajectories. In order to increase the precision of the calculation of the latent-space representations of the trajectory data of the normal trajectories, information of a map representation or information of a roadway condition or information of prevailing traffic regulations or information regarding a prevailing traffic volume or weather conditions or time-of-day and/or day-of-the-week information, which each describe in more detail the conditions under which the trajectory data were recorded during the plurality of trips of the vehicles, can also be taken into account in the training. The additional information can further specify the driving behavior of the vehicles to be respectively classified as normal. Taking into account the various categories of information is based on the idea that the respective driving behavior can turn out differently depending on the traffic volumes or weather conditions, road surfaces or times of day or days of the week. Taking them into account thus enables a precise definition of the normal driving behavior and enables the autoencoder architecture to classify more accurately and create the respective latent-space representations of the recorded trajectory data more precisely. Upon completion of the training, the encoder module of the autoencoder architecture is better configured to generate a corresponding latent-space representation, whereby the corresponding classification module is better configured to carry out the classification of the driving behavior of the road users arranged in the environment of the ego vehicle.

According to one embodiment of the present invention, a normal driving behavior of the road user comprises:

- driving in a lane, designated according to prevailing traffic regulations, of the traveled roadway; and/or
- driving according to prevailing speed regulations;
- following a path of the roadway; and/or driving in a direction of travel designated by prevailing traffic regulations for the respectively traveled roadway and/or lane; and wherein an abnormal driving behavior comprises:
- not following the path of the roadway; and/or driving in a meandering direction of travel deviating from a straight direction of travel; and/or
- driving in a direction of travel opposite the direction of travel designated according to the prevailing traffic regulation; and/or
- driving at an excessive speed; and/or
- driving in a lane not designated according to the prevailing traffic regulations.

This can achieve the technical advantage that precise classification of the driving behavior of the further road users is enabled.

According to one embodiment of the present invention, the received trajectory data of the further road user in the environment of the ego vehicle are based on environmental sensor data of at least one environmental sensor of the ego vehicle, wherein the environmental sensor data comprise camera data, LiDAR data, RADAR data, and/or acoustic data.

This can achieve the technical advantage that precise ascertainment of the trajectory data of the driving trajectories of the road users arranged in the environment of the ego vehicle is enabled. This allows a precise classification of the driving behavior of the respective road users.

According to a further aspect of the present invention, a method for controlling a vehicle is provided. According to an example embodiment of the present invention, the method comprises:

carrying out a classification of a driving behavior of a road user in the environment of an ego vehicle based on trajectory data of the road user according to the method for classifying a driving behavior of a road user of an ego vehicle according to one of the above-described embodiments; and

controlling the ego vehicle, taking into account the classification of the driving behavior of the further road user.

This can achieve the technical advantage that an improved method for controlling an ego vehicle can be provided. By carrying out the method for classifying a driving behavior of a road user of an ego vehicle, an improved classification can be achieved with the aforementioned technical advantages.

According to one embodiment of the present invention, controlling the ego vehicle comprises:

carrying out a safety maneuver if the driving behavior of the further road user is classified as an abnormal driving behavior.

This can achieve the technical advantage that a corresponding response to the classified driving behavior of the road user is enabled and allows safe driving of the ego vehicle. By carrying out the safety maneuver, it is possible to respond to abnormal driving behavior of further road users in order to be able to ensure the safety of the ego vehicle.

According to one embodiment of the present invention, the safety maneuver comprises:

reducing a speed of the ego vehicle; and/or
increasing a distance of the ego vehicle to the further road user; and/or
changing a lane, traveled by the ego vehicle, of the traveled roadway.

This can achieve the technical advantage that it is possible to carry out effective safety maneuvers that can ensure a safety of the ego vehicle. The safety of the ego vehicle within the meaning of the application relates to a collision-free trip of the ego vehicle without collision with the further road user. The safety maneuver to be carried out can in particular be selected, taking into account the respectively classified abnormal behavior of the further road user.

According to a further aspect of the present invention, a computing unit is provided and configured to carry out the method for classifying a driving behavior of a road user in the environment of an ego vehicle according to one of the above-described embodiments and/or the method for controlling an ego vehicle according to one of the above-described embodiments.

According to a further aspect of the present invention, a computer program product is provided and comprises instructions that, when the program is executed by a data processing unit, cause the latter to carry out the method for classifying a driving behavior of a road user in the environment of an ego vehicle according to one of the above-described embodiments and/or the method for controlling an ego vehicle according to one of the above-described embodiments.

Exemplary embodiments of the present invention are explained with reference to the figures.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 shows a schematic representation of a system for classifying a driving behavior of a road user in the environ-

ment of an ego vehicle, according to an example embodiment of the present invention.

FIG. 2 shows a graphical representation of a classification of a driving behavior, according to an example embodiment of the present invention.

FIG. 3 shows schematic representations of two driving situations of an ego vehicle.

FIG. 4 shows a graphical representation of trajectory data.

FIG. 5 shows a graphical representation of latent-space representations of the trajectory data in FIG. 4.

FIG. 6 shows a further graphical representation of latent-space representations of the trajectory data in FIG. 4.

FIG. 7 shows a flow chart of a method for trajectory planning for an ego vehicle, according to an example embodiment of the present invention.

FIG. 8 shows a flow chart of a method for controlling an ego vehicle, according to an example embodiment of the present invention.

FIG. 9 shows a schematic representation of a computer program product.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

FIG. 1 shows a schematic representation of a system 300 for classifying a driving behavior of a road user 303 in the environment 305 of an ego vehicle 301.

FIG. 1 shows a traffic situation with an ego vehicle 301 driving on a roadway 302. A further road user 303 is arranged, as a further vehicle in the embodiment shown, in the environment 305 of the ego vehicle 301. The ego vehicle 301 comprises a computing unit 307 with a classification module 309. The classification module 309 is configured to carry out the method according to the present invention for classifying a driving behavior of a further road user 303 in the environment 305 of an ego vehicle 301.

For classifying a driving behavior of the road user 303, trajectory data 311 are first received by the computing unit 307 or the classification module 309. The trajectory data 311 describe a driving trajectory of the road user 303 and comprise at least position information of the road user 303 at various time points. The trajectory data 311 may additionally comprise velocity information of the road user 303.

The ego vehicle 301 furthermore comprises at least one environmental sensor 313 by means of which the environment 305 of the ego vehicle 301 can be perceived. The trajectory data 311 may thus be given by environmental sensor data of the environmental sensor 313; the driving trajectory carried out by the road user 303 can accordingly be detected by the environmental sensors 313 of the ego vehicle 301. The environmental sensor data may comprise LiDAR data, radar data, camera data, and/or acoustic data of corresponding acoustic sensors.

For classifying the driving behavior, the classification module 309 is configured to calculate latent-space representations 325 for the received trajectory data 311 of the road user 303. The latent-space representations 325 describe reduced-dimension representations of the trajectory data 311. The latent-space representations 325 thus represent representations of the corresponding trajectory data 311 in a latent space 333 and may be designed, for example, as vector representations of the trajectory data 311.

In the embodiment shown, the classification module 309 comprises, for this purpose, an encoder module 317 of an autoencoder architecture 315 shown simplified in graph a).

The autoencoder architecture 315 comprises a corresponding decoder module 319 in addition to the encoder

module 317. The autoencoder architecture 315 is configured to calculate latent-space representations 325, corresponding to the input data 321, in the latent space. Since the latent-space representations 325 of the input data 321 at least largely comprise the information of the input data 321 but are reduced in dimensionality in comparison to the input data 321, the latent-space representations 325 represent an encoding of the information of the input data 321. The decoder module 319 of the autoencoder architecture 315 is furthermore configured to reproduce the input data 321 and generate corresponding reproduced input data 323 based on the latent-space representations 325 of the input data 321.

By correspondingly training the autoencoder architecture 315, corresponding latent-space representations 325 can thus be created for given input data 321.

According to the present invention, the autoencoder architecture 315 is thus trained based on trajectory data 327. The trajectory data 327 are based on a plurality of trips of a vehicle or a plurality of vehicles and describe the driving trajectories respectively carried out by the vehicles. The training of the autoencoder architecture 315 may furthermore be carried out, taking into account map data 329 comprising map information of the roadways traveled by the respective vehicles for the purpose of recording the trajectory data 327. Furthermore, information regarding the roadway condition, the traffic volume, regarding weather conditions, regarding times of day or days of the week as well as regarding prevailing traffic regulations that existed during the trips for recording the trajectory data 327 can be taken into account in the training. The corresponding trajectory data 327, map data 329, or data of said information 331 can be taken into account in the shown input data 321 of the autoencoder architecture 315.

After corresponding training of the autoencoder architecture 315, the respective encoder module 317 is configured to calculate corresponding latent-space representations 325 in the latent space for the received trajectory data 311 of the driving trajectory carried out by the road user 303.

During the training of the autoencoder architecture 315, trajectory data 327 of driving trajectories based on a normal driving behavior of the respective vehicles are primarily used. The normal driving behavior is characterized in that prevailing traffic rules are taken into account, in that the respective vehicle follows the path of the traveled roadway evenly, in that no jerky or unforeseeable steering movements or maneuvers are carried out, and in that there is, for example, an even velocity profile. The normal driving behavior can be adjusted to the respective weather conditions, the present traffic volume, the time of day or the day of the week as well as to the respective roadway conditions or the prevailing traffic rules. The features of a normal driving behavior mentioned herein are merely exemplary. A normal driving behavior may comprise more or less of the mentioned features, depending on the traffic situation. A limitation of the present invention is not to be achieved thereby.

Driving trajectories of a normal driving behavior, which are called normal trajectories within the meaning of the application, show great similarities to one another and differ only slightly from one another. The corresponding driving trajectories have largely common paths that correspond to the paths of the respective roadway and are also similar to one another in terms of steering movements or maneuvers carried out as well as the respective velocity of the vehicle.

During the training of the autoencoder architecture 315, the encoder module 317 is trained to convert the trajectory data 327 into corresponding latent-space representations

325. The encoder module 317 is trained to convert similar or identical trajectory data 327 into similar or identical latent-space representations 325. Similar or identical latent-space representations 325 have, according to the present invention, only small distances to one another in the latent space and are arranged in common regions in the latent space. Similar or identical latent-space representations 325 describe vector representations whose entries are the same or have only slight deviations from one another.

On the other hand, driving trajectories based on abnormal driving behavior, which, for example, comprises a failure to comply with prevailing traffic regulations, not following the path of the traveled roadway or a presence of jerky steering movements, uneven velocity profiles or, for example, zig-zag or meandering driving behavior, are converted by the appropriately trained encoder module 317 into corresponding latent-space representations 325 that have a relatively large distance in the latent space to the latent-space representations 325 of the normal trajectories.

For classifying the driving trajectories of the road users 303 arranged in the environment 305 of the ego vehicle 301, the classification module 309 is configured to determine distances between latent-space representation 325 of the received trajectory data 311 of the driving trajectories driven by the road users 303, in the latent space to latent-space representations 325 of normal trajectories that were learned during the training process of the autoencoder architecture 315 by the autoencoder architecture 315, and in particular by the encoder module 317.

For this purpose, the classification module 309 comprises a distance module 318 in addition to the encoder module 317. The distance module 318 is configured to determine distances D of the latent-space representations 325 of the received trajectory data 311 of the driving trajectories of the road users 303 to the learned latent-space representations 325 of the normal trajectories.

The distances D may be ascertained by the distance module 318 according to a predetermined distance metric within the latent space. For example, the distance metric may be given by a Euclidean distance function or by a logarithmic likelihood function. By means of the mentioned functions, the distance module 318 can thus ascertain distances between the latent-space representations 325 of the received trajectory data 311 of the driving trajectories of the road users 303 to the latent-space representations 325, learned in the training process, of the normal trajectories based on normal driving behavior.

Based on the ascertained distances D, the classification module 309 classifies the driving behavior of the road users 303 arranged in the environment 305. In so doing, a driving behavior of a road user 303 is classified as a normal driving behavior if the respective latent-space representations 325 of the corresponding driving trajectories of the road user 303 have a distance D to latent-space representations 325 of the normal trajectories in the latent space that is less than a predetermined limit value. On the other hand, if the latent-space representations 325 of the driving trajectories carried out by the respective road user 303 have distances D to the latent-space representations 325 of the normal trajectories in the latent space that are greater than or equal to the predetermined limit value, the corresponding driving behavior of the road user 303 is classified as an abnormal driving behavior.

According to the present invention, classification information is subsequently output, which describes the classification of the driving behavior of the road user 303 arranged in the environment 305.

Based on the classification information, the control of the ego vehicle 301 can be adjusted. For example, a safety maneuver may be carried out or caused if the driving behavior of the road user 303 is classified as an abnormal driving behavior. For example, the corresponding safety maneuver may comprise reducing the speed of the ego vehicle 301 and/or increasing a distance of the ego vehicle 301 to the respective road user 303. Alternatively or additionally, the safety maneuver may comprise a lane change from a traveled lane 306 to a non-traveled lane 304. Alternatively or additionally, further maneuvers may be carried out, which appropriately take into account the driving behavior, correspondingly classified as abnormal driving behavior, of the road user 303 arranged in the environment 305.

According to the present invention, the autoencoder architecture 315, and in particular the encoder module 317, can be designed according to conventional architectures for autoencoders.

According to one embodiment, the autoencoder architecture 315 is designed as a variational autoencoder VAE. In this case, the correspondingly generated latent-space representations 325 are designed as statistical distributions in the latent space and are each characterized by an expected value μ and a variance σ^2 . For example, the statistical distributions of the latent-space representations 325 may be designed as Gaussian normal distributions.

The training process of the autoencoder architecture 315 can in particular be designed as a supervised learning process.

FIG. 2 shows a graphical representation of a classification of a driving behavior.

FIG. 2 shows a graphical and highly simplified representation of the latent space 333 and the respective latent-space representations 335 of the normal trajectories T2 as well as the latent-space representations 334 of the driving trajectories T1 of the road users 303 arranged in the environment 305 of the ego vehicle 301.

During the training process of the autoencoder architecture 315, the corresponding encoder module 317 is trained to generate the calculation of the latent-space representations 335 of the trajectory data 327 used in the training process. The autoencoder architecture 315, and in particular the encoder module 317, learn to convert normal trajectories T2, which represent a normal driving behavior and are highly similar to one another despite having been carried out by various vehicles, into likewise highly similar latent-space representations 335 that are arranged within the latent space 333 at small distances from one another.

After successful training, the encoder module 317 is thus configured to calculate corresponding latent-space representations 334 in the latent space 333 for the observed driving trajectories T1 of the detected road users 303 and, in particular, for the corresponding trajectory data 311. If the driving trajectories T1 of the road users 303 are based on normal driving behavior, the corresponding latent-space representations 334 will be highly similar to the latent-space representations 335 of the normal trajectories T2 and be arranged in the latent space 333 at a small distance to the respective latent-space representations 334 of the normal trajectories T2. On the other hand, if the driving trajectories T1 of the road users 303 are based on abnormal driving behavior, the corresponding driving trajectories T1 will have great differences from the normal trajectories T2 and the correspondingly generated latent-space representations 334 of the driving trajectories T1 will have a corresponding large

distance D to the latent-space representations 335 of the normal trajectories T2 in the latent space 333.

FIG. 2 shows a situation of an abnormal driving behavior of a road user 303. The corresponding driving trajectory T1 has a great difference from the normal trajectories T2, which difference is shown in FIG. 2 by the meandering path of the driving trajectory T1 in comparison to the straight curved path of the normal trajectory T2. The correspondingly generated latent-space representation 334 of the driving trajectory T1 thus has a relatively large distance D to the latent-space representations 335 of the normal trajectories T2.

FIG. 2 shows three latent-space representations 335 for three different normal trajectories T2 in the latent space 333. This is only exemplary. In reality, a plurality of different trajectories T2 can be ascertained, which are all based on normal driving behavior, differ from one another but have a large number of commonalities due to the underlying normal driving behavior, and are thus arranged in a comparatively narrow area within the latent space 333. In FIG. 2, the distances between the latent-space representations 334, 335 are not to scale so that the distance D of the latent-space representation 334 to the latent-space representations 335 may appear to be less than the distances between the latent-space representations 335. This is unintended and is owed to the illustration.

According to the present invention, a predetermined limit value is defined for the classification of the driving behavior of the road users 303 and in particular for the distance determination of the distances D of the latent-space representations 334 of the driving trajectories T1 to the latent-space representations 335 of the normal trajectories T2. This limit value can be individually adjusted depending on various criteria, for example the degree of similarity of the normal trajectories T2 for different driving situations.

A correspondingly greater limit value for the distance determination can be selected for driving situations in which the normal trajectories T2 have great differences from one another due to the present driving situation, even though all are based on a normal driving behavior. For driving situations in which the normal trajectories T2 have only slight differences from one another and the corresponding latent-space representations 335 of the individual normal trajectories T2 in the latent space 333 are arranged in a correspondingly smaller area with smaller distances to one another, the predetermined limit value can accordingly be provided with a lower numerical value so that even smaller distances of the latent-space representation 334 of the driving trajectories T1 of the road users 303 to the latent-space representations 335 of the normal trajectories T2 already result in the classification of a normal driving behavior of the respective road users 303.

In the embodiment shown, the autoencoder architecture 315 is designed as a variational autoencoder VAE. The corresponding latent-space representations 335 are formed by statistical distributions 337 within the latent space 333. The statistical distributions 337 are characterized by expected values μ and variances σ^2 and are represented in FIG. 2 by the circular structures. The individual latent-space representations 335 thus have corresponding flat formations in the latent space 333 according to the statistical distributions 337 so that the latent space 333 is embodied by a continuous occupancy of the latent-space representations.

As already described with respect to FIG. 1, the classification module 309 comprises, in addition to the encoder module 317, the distance module 318 already described. The distance module 318 is configured to ascertain the distances

D between the latent-space representations **334** of the driving trajectories T1, ascertained via the received trajectory data **311**, of the road users **303** arranged in the environment **305** of the ego vehicle **301**, to the latent-space representations **335**, learned in the training process of the autoencoder architecture **315**, of the normal trajectories T2 based on normal driving behavior. As already described, this can take place by carrying out a corresponding distance metric, which may be designed, for example, as a Euclidean distance function or a logarithmic likelihood function.

FIG. **3** shows schematic representations of two driving situations of an ego vehicle **301**.

The two graphs a) and b) of FIG. **3** show two different driving situations with an ego vehicle **301** and a further road user **303**.

In both driving situations, both the ego vehicle **301** and the further road user **303** drive in a lane **302**. The ego vehicle **301** and the further road user **303** drive in opposite directions of travel in the driving situations shown.

In graph a), the further road user **303** shows a normal driving behavior. The latter is characterized by the relatively straight driving trajectory T1 of the road user **303**, the path of which largely follows the path of the traveled roadway **302**. The ego vehicle **301** likewise shows a normal driving behavior, which in turn is represented by the relatively straight driving trajectory T3, which in turn follows the path of the traveled roadway **302**.

On the other hand, the road user **303** in graph b) shows an abnormal driving behavior. This is represented by the more or less meandering driving trajectory T2 of the road user **303**.

FIG. **4** shows a graphical representation of trajectory data **327**.

FIG. **4** shows a multitude of different trajectory data **327**, **328** of a plurality of different driving trajectories along a lane **302**. The lane **302** shown is in the form of a roundabout. The individual driving trajectories each have **25** data points and comprise a time period of 5 measurement seconds. In the following, only a part of the driving trajectories shown are taken into account, in particular the driving trajectories starting from the marking M shown in the clockwise direction along the roundabout or the first entry to the roundabout.

FIG. **5** shows a graphical representation of latent-space representations **325** of the trajectory data **327** in FIG. **4**.

Graph a) shows a representation in the latent space of latent-space representations **325**, **326** of the trajectory data **328** shown in FIG. **4**, starting from marking M. The latent space shown has two spatial dimensions DM1, DM2. However, this is only exemplary. According to the present invention, the latent space may have any number of spatial dimensions, depending on the representation.

In graph a), the latent-space representations **326** of the trajectory data **328** have a high spread, while the latent-space representations **325** of the trajectory data **327** are arranged in a relatively small space region. Graph b) shows an enlarged representation of the space region in which the latent-space representations **325** are arranged. In this space region, the latent-space representations **326** of the trajectory data **328** are likewise arranged closely next to the latent-space representations **325**. A differentiation of the latent-space representations **325**, **326** is not possible here.

FIG. **6** shows a further graphical representation of latent-space representations **325** of the trajectory data in FIG. **4**.

Graph a) shows a graphical representation of the trajectory data **327**, **328** from FIG. **5**. In the representation shown, the individual trajectory data **327**, **328** are provided with an

additional drift value. This allows a clear separation of the trajectory data from one another to be visible.

Graph b) shows a graphical representation in the latent space of the latent-space representations **325**, **326** of the trajectory data **327**, **328** from graph a). The latent space shown here has three spatial dimensions DM1, DM2, DM3. As a result of the additional spatial dimension DM3, a clear separation of the two different latent-space representations **325**, **326** to one another can be seen.

FIG. **7** shows a flow chart of a method **100** for classifying a driving behavior of a road user in the environment of an ego vehicle **301**.

According to the present invention, in a first method step **101**, trajectory data **311** of a road user **303** arranged in an environment **305** of an ego vehicle **301** are received. The trajectory data **311** comprise at least one position information item of the driving trajectory T1 followed by the road user **303**.

For example, the trajectory data may be based on environmental sensor data of at least one environmental sensor **313** of the ego vehicle **301**. The environmental sensor data may be given, for example, by camera data, LiDAR data, radar data, or acoustic data.

In a further method step **103**, a latent-space representation **334** of the driving trajectory T1 in the latent space is ascertained based on the trajectory data **311**. The latent-space representation **334** can be generated by an appropriately trained encoder module **317** of an autoencoder architecture **315**. According to one embodiment, the autoencoder architecture can be designed as a variational autoencoder VAE. The corresponding latent-space representations **334** can accordingly be designed as statistical distributions in the latent space **333** and be characterized by expected values μ and variances σ^2 . For example, the statistical distributions may be designed as Gaussian normal distributions.

In a further method step **105**, a distance D of the latent-space representation **334** of the driving trajectory T1 of the road user **303** to a latent-space representation **335** of at least one normal trajectory T2 in the latent space **333** is ascertained. For this purpose, an appropriately trained distance module **318** is applied and configured to ascertain, by carrying out a corresponding distance metric, distances D between the latent-space representations **334** of the detected driving trajectories T1 of the road users **303** to latent-space representations **335**, learned in the training process of the autoencoder architecture **315**, of normal trajectories T2. For distance determination, the distance module **318** can apply a corresponding distance metric, which is, for example, designed as a Euclidean distance function or a logarithmic likelihood function.

In a further method step **107**, the driving behavior of the road user **303** is classified as a normal driving behavior if the distance D of the latent-space representation **334** of the driving trajectory T1 of the road user **303** to the latent-space representation **335** of the normal trajectory T2 in the latent space **333** falls below a predetermined limit value (see step **106**).

In a method step **109**, on the other hand, the driving behavior of the road user **303** is classified as an abnormal driving behavior if the distance D of the latent-space representation **334** of the driving trajectory T1 of the road user **303** to the latent-space representation **335** of the normal trajectory T2 in the latent space **333** exceeds the predetermined limit value (see step **106**).

In a further method step **111**, corresponding classification information is output, which describes the classification of the driving behavior of the road user **303**.

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FIG. 8 shows a flow chart of a method 200 for controlling an ego vehicle 301.

According to the present invention, in a first method step 201, a classification of a driving behavior of a further road user located in an environment of the ego vehicle 301 is carried out by carrying out the method 100 according to the present invention for classifying a driving behavior of a road user in the environment of an ego vehicle 301 according to the embodiments described above.

In a further method step 203, the ego vehicle 301 is controlled according to the respective planned trajectory.

For this purpose, a safety maneuver is carried out in a further method step 205 if the driving behavior of the further road user is classified as an abnormal driving behavior.

The safety maneuver may comprise reducing a speed of the ego vehicle 301 and/or increasing a distance of the ego vehicle 301 to the further road user and/or changing a lane, traveled by the ego vehicle, of the traveled roadway. Alternatively or additionally, further maneuvers may be carried out, by means of which a safety of the ego vehicle 301 can be increased and, in particular, a collision with the further road user can be avoided.

Controlling the ego vehicle 301 comprises, in particular, outputting corresponding control signals from the computing unit to actuators of the ego vehicle 301 to carry out the safety maneuver.

FIG. 9 shows a schematic representation of a computer program product 400 comprising instructions that, when the program is executed by a computing unit, cause the latter to carry out the method 100 for classifying a driving behavior of a road user in the environment of an ego vehicle 301 and/or the method 200 for controlling an ego vehicle 301.

The computer program product 400 is stored on a storage medium 401 in the embodiment shown. The storage medium 401 can be any storage medium available in the related art.

What is claimed is:

1. A method for controlling an ego vehicle, comprising the following steps:

carrying out a classification of a driving behavior of a road user in an environment of the ego vehicle based on trajectory data of a driving trajectory of the road user by:

receiving trajectory data of a driving trajectory of the road user arranged in the environment of the ego vehicle, wherein the trajectory data include at least position information of the driving trajectory followed by the road user,

ascertaining a latent-space representation of the driving trajectory of the road user in a latent space, wherein the latent-space representation of the driving trajectory is given by a reduced-dimension representation of the trajectory data,

ascertaining a distance between the latent-space representation of the driving trajectory of the road user and a latent-space representation of at least one normal trajectory in the latent space, wherein the normal trajectory is classified as a driving trajectory based on a normal driving behavior,

classifying whether a driving behavior of the road user is a normal driving behavior or an abnormal driving behavior, according to a condition that the normal driving behavior classification is applied based on the distance between of the latent-space representation of the driving trajectory of the road user to and the latent-space representation of the at least one normal trajectory in the latent space falling below a predetermined limit value, and the abnormal driving

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behavior classification is applied based on the distance of the latent-space representation of the driving trajectory of the road user to the latent-space representation of the at least one normal trajectory in the latent space exceeding the predetermined limit value, and

outputting classification information; and
controlling the ego vehicle taking into account the classification of the driving behavior of the road user.

2. The method according to claim 1, wherein the distance is determined by a predetermined distance metric, and wherein the distance metric is given by a Euclidean distance function or by a logarithmic likelihood function of the latent-space representations within the latent space.

3. The method according to claim 1, wherein:
the ascertainment of the latent-space representation is carried out by a classification module that includes an encoder module of an autoencoder architecture that is trained to ascertain latent-space representations of driving trajectories, including the latent-space representation of the driving trajectory of the road user, in the latent space;

the classifying of whether the driving behavior is the normal driving behavior or the abnormal driving behavior is performed by the classification module;

the encoder module is trained to recognize and classify that the latent-space representation corresponds to a driving trajectory that is valid for the control of the ego vehicle based on the normal driving behavior;

the autoencoder architecture is configured as a variational autoencoder; and

the latent-space representations of the driving trajectories are statistical distributions in the latent space and are each characterized by an expected value μ and a variance σ^2 .

4. The method according to claim 3, wherein the distributions are Gaussian normal distributions in the latent space.

5. The method according to claim 1, wherein:

the ascertainment of the latent-space representation is carried out by a classification module that includes an encoder module of an autoencoder architecture that is trained to ascertain latent-space representations of driving trajectories, including the latent-space representation of the driving trajectory of the road user, in the latent space;

the classifying of whether the driving behavior is the normal driving behavior or the abnormal driving behavior is performed by the classification module;

the encoder module is trained to recognize and classify that the latent-space representation corresponds to a driving trajectory that is valid for the control of the ego vehicle based on the normal driving behavior; and

the training of the autoencoder architecture is carried out on trajectory data of a plurality of trips of a vehicle or a plurality of vehicles along a roadway traveled by the ego vehicle; and

map information of a map representation mapping the respectively traveled roadway and/or information regarding a roadway condition and/or information regarding prevailing traffic regulations and/or information regarding prevailing traffic volumes and/or information regarding prevailing weather conditions and/or time-of-day and/or day-of-the-week information, are also taken into account in the training.

6. The method according to claim 1, wherein:
the normal driving behavior of the road user includes:

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driving in a lane, designated according to prevailing traffic regulations, of a traveled roadway; and/or driving according to prevailing speed regulations; following a path of the traveled roadway; and/or driving in a direction of travel designated by prevailing traffic regulations for the traveled roadway and/or lane; and

the abnormal driving behavior includes:

not following a path of the roadway; and/or driving in a meandering direction of travel deviating from a straight direction of travel; and/or driving in a direction of travel opposite a direction of travel designated according to the prevailing traffic regulation; and/or driving at an excessive speed; and/or driving in a lane not designated according to the prevailing traffic regulations.

7. The method according to claim 1, wherein the received trajectory data of the road user in the environment of the ego vehicle are based on environmental sensor data of at least one environmental sensor of the ego vehicle, and wherein the environmental sensor data include camera data, and/or LiDAR data, and/or RADAR data, and/or acoustic data.

8. The method according to claim 1, wherein the controlling of the ego vehicle includes:

carrying out a safety maneuver based on the driving behavior of the road user being classified as an abnormal driving behavior.

9. The method according to claim 8, wherein the safety maneuver includes:

reducing a speed of the ego vehicle; and/or increasing a distance of the ego vehicle to the road user; and/or changing a lane, traveled by the ego vehicle, of a traveled roadway.

10. A computing unit comprising a processor system that includes at least one processor that is programmed to:

receive trajectory data of a driving trajectory of a road user arranged in an environment of an ego vehicle, wherein the trajectory data include at least position information of the driving trajectory followed by the road user;

ascertain a latent-space representation of the driving trajectory of the road user in a latent space, wherein the latent-space representation of the driving trajectory is given by a reduced-dimension representation of the trajectory data;

ascertain a distance between the latent-space representation of the driving trajectory of the road user and a latent-space representation of at least one normal trajectory in the latent space, wherein the normal trajectory is classified as a driving trajectory based on a normal driving behavior;

classify whether a driving behavior of the road user is a normal driving behavior or an abnormal driving behav-

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ior, according to a condition that the normal driving behavior classification is applied based on the distance between the latent-space representation of the driving trajectory of the road user and the latent-space representation of the at least one normal trajectory in the latent space falling below a predetermined limit value, and the abnormal driving behavior classification is applied based on the distance of the latent-space representation of the driving trajectory of the road user to the latent-space representation of the at least one normal trajectory in the latent space exceeding the predetermined limit value;

output classification information; and

control the ego vehicle taking into account the classification of the driving behavior of the road user.

11. A non-transitory computer-readable medium in which is stored a computer program including instructions for classifying a driving behavior of a road user in an environment of an ego vehicle, the instructions, when executed by a data processor, causing the data processor to perform the following steps:

receiving trajectory data of a driving trajectory of the road user arranged in the environment of the ego vehicle, wherein the trajectory data include at least position information of the driving trajectory followed by the road user;

ascertaining a latent-space representation of the driving trajectory of the road user in a latent space, wherein the latent-space representation of the driving trajectory is given by a reduced-dimension representation of the trajectory data;

ascertaining a distance between the latent-space representation of the driving trajectory of the road user and a latent-space representation of at least one normal trajectory in the latent space, wherein the normal trajectory is classified as a driving trajectory based on a normal driving behavior;

classifying whether a driving behavior of the road user is a normal driving behavior or an abnormal driving behavior, according to a condition that the normal driving behavior classification is applied based on the distance between the latent-space representation of the driving trajectory of the road user and the latent-space representation of the at least one normal trajectory in the latent space falling below a predetermined limit value, and the abnormal driving behavior classification is applied based on the distance of the latent-space representation of the driving trajectory of the road user to the latent-space representation of the at least one normal trajectory in the latent space exceeding the predetermined limit value;

outputting classification information; and

controlling the ego vehicle taking into account the classification of the driving behavior of the road user.

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